

Macroeconomics 1

Workers and Capitalists

- (a) The marginal product of capital is $f'(k)$. The share of workers and capitalist would thus be $f'(k)k_w, f'(k)k_c$. Hence, the aggregate saving is

$$s_{ww}(f(k) - kf'(k)) + s_{wp}k_wf'(k) + s_cf'(k)k_c \quad (1)$$

Hence, G is:

$$G = \begin{bmatrix} G_w(k_w, k_c) \\ G_c(k_w, k_c) \end{bmatrix} = \frac{1}{1+n} \begin{bmatrix} s_{ww}(f(k) - kf'(k)) + s_{wp}f'(k)k_w + (1-\delta)k_w \\ s_cf'(k)k_c + (1-\delta)k_c \end{bmatrix} \quad (2)$$

- (b) At $k = k^*$, $G(k^*) = k^*$. We would have:

$$(n + \delta)k_w = s_{ww}(f(k) - kf'(k)) + s_{wp}k_wf'(k) \quad (3)$$

$$(n + \delta)k_c = s_cf'(k)k_c \quad (4)$$

At $k = 0$, for equations (3), (4), RHS equals LHS equals 0. This would be valid if $\alpha = 0$ (Check Commendatore et al. [2005] for further details).

- (c) Equilibrium (i) is also known as Pasinetti equilibrium (Pasinetti [1962]). Since $k_c \neq 0$, using (4) gives $f'(k) = \frac{n+\delta}{s_c}$. Substitute into LHS of (3) to obtain:

$$\begin{aligned} s_cf'(k)k_w &= s_{ww}(f(k) - kf'(k)) + s_{wp}k_wf'(k) \Rightarrow \\ k_w &= \frac{s_{ww}}{s_c - s_{wp}} \left[\frac{f(k)}{kf'(k)} - 1 \right] k \Rightarrow \\ k_w &= \frac{s_{ww}}{s_c - s_{wp}} \left[\frac{1 - \epsilon}{\epsilon} \right] k \end{aligned} \quad (5)$$

Which in turn gives:

$$k_c = k - k_w = \left[1 - \frac{s_{ww}}{s_c - s_{wp}} \frac{1 - \epsilon}{\epsilon} \right] k \quad (6)$$

- (d) Equilibrium (ii) is called dual equilibrium. Assume $k_c = 0$. So $k = k_w$ and (3) would give

$$\begin{aligned} (n + \delta)k &= s_{ww}(f(k) - kf'(k)) + s_{wp}kf'(k) \Rightarrow \\ (n + \delta) \frac{k}{f(k)} &= s_{ww}(1 - \epsilon) + s_{wp}\epsilon \Rightarrow \\ \frac{f(k)}{k} &= \frac{n + \delta}{s_{ww}(1 - \epsilon) + s_{wp}\epsilon} \end{aligned} \quad (7)$$

(e) Denote J_G as the Jacobian of G :

$$J_G = \begin{bmatrix} \frac{\partial G_w}{\partial k_w} & \frac{\partial G_w}{\partial k_c} \\ \frac{\partial G_c}{\partial k_w} & \frac{\partial G_c}{\partial k_c} \end{bmatrix} = \frac{1}{1+n} \begin{bmatrix} (1-\delta) + (s_{wp} - s_{ww})kf''(k) + s_{wp}(f'(k) - k_cf''(k)) & (s_{wp} - s_{ww})kf''(k) - s_{wp}k_cf''(k) \\ s_ck_cf''(k) & (1-\delta) + s_cf'(k) + s_ck_cf''(k) \end{bmatrix} \quad (8)$$

For trivial fixed point, if it exists, the Jacobian is:

$$J^{iii} = \frac{1}{1+n} \begin{bmatrix} (1-\delta) + s_{wp} & 0 \\ 0 & (1-\delta) + s_c \end{bmatrix} \quad (9)$$

It's natural to assume that $s \gg \delta, n$. Hence, $\lambda_{1,2} > 1$, which in turn implies that it's a source.

For dual equilibrium fixed point, the Jacobian is:

$$J^{ii} = \frac{1}{1+n} \begin{bmatrix} (1-\delta) + (s_{wp} - s_{ww})kf''(k) + s_{wp}kf'(k) & (s_{wp} - s_{ww})kf''(k) \\ 0 & (1-\delta) + s_ckf'(k) \end{bmatrix} \quad (10)$$

Its Eigenvalues are the diagonal entries. The type of the Eigenvalues are ambiguous. Consider the simple case where $s_{wp} = s_{ww}$. This would imply that $(1+n)\lambda_1 = (1-\delta) + s_{wp}kf'(k)$ and $(1+n)\lambda_2 = (1-\delta) + s_ckf'(k)$. Since $s_c > s_{wp}$, the condition for stability would be $\lambda_2 < 1$. This requires

$$\epsilon f < \frac{n+\delta}{s_c}$$

which can hold for some parameter values, even though it's very unlikely to be so.

Assume η is elasticity of marginal product. The Eigenvalues of type (i) fixed points are:

$$\lambda_{1,2} = \frac{1}{2}X \pm \sqrt{X^2 - 4Y} \quad (11)$$

where

$$X = \frac{n+\delta}{1+n} \left[\frac{2(1-\delta)}{n+\delta} + 1 + \frac{s_{wp}\epsilon - s_{ww}\eta}{s_c\eta} + \eta \right]$$

$$Y = \frac{1-\delta}{1+n} X - \left(\frac{1-\delta}{1+n} \right)^2 + \frac{\eta(s_{wp} - s_{ww}) + s_{wp}}{s_c} \left(\frac{n+\delta}{1+n} \right)^2$$

The type of this zero is, yet again, unambiguous. In fact, only the trivial fixed point's type, even if we assume $f = 0$ if $k = 0$, regardless of the value of α , is clear. Since $s_c > s_{wp}$, for it to be stable, we need

$$f'(0) < \frac{\delta+n}{s_c}$$

But we know that f is concave, so f' is increasing in k . On the other hand, $f'(k^i) = \frac{n+\delta}{s_c}$, but $k^i > k^{iii} = 0$, hence $f'(0)$ is always greater than $\frac{\delta+n}{s_c}$.

References

- P. Commendatore et al. Complex dynamics in a pasinetti–solow model of growth and distribution. *Macroeconomics*, 511006, 2005.
- L. L. Pasinetti. Rate of profit and income distribution in relation to the rate of economic growth. *The Review of Economic Studies*, 29(4):267–279, 1962.